

LARISSA, Greece

WMO: 166480

Lat: 39.650N

Lon: 22.466E

Elev: 241

StdP: 14.57

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	23.4	26.8	15.3	12.0	32.2	19.3	14.7	33.8	23.1	42.3	20.1	43.3	1.5	300	0.426

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.6	98.5	69.5	95.3	69.0	93.1	68.5	72.9	90.4	71.6	88.9	70.4	87.0	8.0	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.2	104.5	76.8	66.5	98.4	76.1	65.3	94.2	75.6	36.7	90.2	35.5	88.8	34.5	87.1	83.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.2	16.6	14.1	DB	16.8	105.2	6.9	3.7	11.9	107.8	7.9	110.0	4.0	112.0	-1.0	114.7	(4)
(5)			WB	16.0	76.5	6.6	2.7	11.3	78.4	7.4	80.0	3.7	81.5	-1.1	83.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	60.4	41.7	44.9	50.0	56.9	66.8	75.8	79.8	78.9	71.3	61.9	52.1	43.6		
	DBStd	14.58	6.26	6.05	5.97	5.33	5.28	4.96	3.96	3.60	5.09	5.57	6.27	6.75		
	HDD50	772	263	160	74	9	0	0	0	0	0	1	48	217		
	HDD65	3230	722	564	467	246	42	0	0	0	9	127	388	665		
	CDD50	4566	6	17	73	217	522	773	924	897	639	369	111	18		
	CDD65	1548	0	0	1	4	99	324	459	432	198	30	1	0		
	CDH74	18266	0	0	23	123	1265	3890	5731	5055	1867	302	10	0		
	CDH80	8867	0	0	7	18	408	1915	3085	2638	735	61	0	0		
(14) Wind	WSAvg	3.7	2.6	3.8	4.3	4.2	4.4	4.8	4.8	4.4	4.0	2.8	2.3	2.4		
(15) Precipitation	PrecAvg	16.7	1.3	1.4	1.5	1.2	1.5	1.0	0.7	0.6	1.3	1.9	2.1	2.0		
	PrecMax	27.7	4.1	4.8	4.6	4.8	5.2	4.2	5.2	2.8	9.4	5.0	6.7	6.1		
	PrecMin	8.3	0.0	0.2	0.4	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.1		
	PrecStd	3.6	1.0	1.0	0.9	1.0	1.1	1.0	1.0	0.7	1.6	1.4	1.5	1.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.3	68.4	76.0	82.6	92.7	100.8	104.3	100.8	95.4	85.9	75.2	66.1		
		MCWB	54.3	54.5	56.8	61.5	65.6	69.6	71.1	69.8	67.0	65.3	60.3	57.0		
	2%	DB	58.7	63.7	71.2	77.4	87.7	96.7	98.7	97.2	91.3	80.8	69.6	60.8		
		MCWB	50.5	52.6	55.4	60.0	64.1	68.5	69.3	69.6	66.9	63.3	59.3	54.6		
	5%	DB	55.2	60.2	67.4	73.5	84.4	93.2	95.4	94.7	87.8	77.1	65.8	56.9		
		MCWB	48.9	51.4	54.3	58.4	63.1	68.0	69.1	69.6	66.0	62.5	57.5	51.8		
	10%	DB	51.7	56.4	63.0	69.9	81.1	89.9	93.0	91.8	84.3	73.4	62.4	53.6		
		MCWB	47.3	49.3	52.3	56.6	62.1	67.2	68.6	68.8	65.3	61.2	56.0	49.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.7	57.0	58.9	64.1	68.1	74.2	74.2	75.6	71.0	68.8	63.0	58.9		
		MCDB	61.5	63.6	72.2	78.2	84.7	93.5	93.2	91.2	83.5	79.0	69.7	62.5		
	2%	WB	51.9	54.0	56.7	61.4	66.1	71.7	72.5	73.3	69.4	66.0	60.6	55.5		
		MCDB	56.9	61.2	68.4	73.9	82.7	91.0	90.6	90.1	82.7	75.0	66.7	59.8		
	5%	WB	49.8	52.0	54.8	59.6	64.8	69.9	71.2	71.8	68.1	64.5	58.9	52.6		
		MCDB	53.6	58.5	65.3	70.8	80.1	87.5	89.1	88.1	81.1	72.4	63.9	56.0		
	10%	WB	47.9	50.0	52.8	57.6	63.5	68.5	70.2	70.5	67.0	63.0	57.0	50.3		
		MCDB	51.1	55.5	62.0	67.4	77.3	85.0	87.7	86.0	79.4	70.3	61.3	52.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	16.5	18.5	21.3	22.9	25.3	26.1	25.6	25.6	23.5	20.5	17.9	15.6		
		MCDBR	23.5	26.4	30.1	30.3	31.1	31.0	30.4	30.1	29.5	27.5	22.7	20.7		
		MCWBR	16.2	16.4	16.6	15.1	11.7	9.7	9.6	9.9	10.6	12.6	13.3	14.3		
	5% WB	MCDBR	20.0	23.6	27.4	26.8	27.5	27.5	26.6	25.8	24.7	21.6	19.2	17.5		
		MCWBR	14.5	15.2	15.9	14.3	11.2	9.9	9.2	9.7	9.7	10.6	12.1	12.7		
(40) Clear-Sky Solar Irradiance	taub		0.360	0.390	0.437	0.490	0.479	0.468	0.475	0.486	0.464	0.428	0.393	0.367		
	taud		2.360	2.231	2.099	1.975	2.073	2.151	2.140	2.101	2.138	2.218	2.269	2.320		
	Ebn at Noon		251	258	257	251	257	259	256	249	246	241	234	236		
	Edh at Noon		28	36	46	56	52	48	48	49	44	37	31	27		
(44) All-Sky Solar Radiation	RadAvg		593	835	1221	1612	1939	2210	2297	2040	1492	995	670	519		
	RadStd		91	106	123	164	99	134	95	92	112	113	79	86		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	+0.68	N/A	N/A	+1.30	N/A	N/A	N/A	-196	N/A	N/A
Regional (10 neighbors)	+0.65	N/A	N/A	+1.19	N/A	N/A	N/A	-160	+184	N/A

Nomenclature: See separate page